4.1: Vectors in $\mathbb{R}^n$

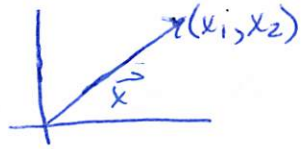
MATH 2360

Definitions

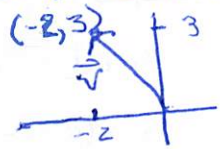
1. **Vector in the Plane:** Geometrically, a vector in the plane is represented by a directed line segment with its initial point at the origin and its terminal point at (x_1, x_2). This ordered pair is used to represent the vector, i.e.,

   $$\vec{x} = (x_1, x_2)$$

   The coordinates x_1, x_2 are the components of vector $\vec{x}$.



Ex: $\vec{v} = (-2, 3)$



2. **Vector Equality:** Two vectors $\vec{u}$ and $\vec{v}$ are equal iff $u_1 = v_1$ and $u_2 = v_2$.

3. **Vector Addition:** To add two vectors in the plane, add their corresponding components:

   $$\vec{u} + \vec{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$$



Ex:

1) Let $\vec{u} = (2, -5)$ and $\vec{v} = (-1, 3)$. Find $\vec{u} + \vec{v}$.

$\vec{u} + \vec{v} = (2, -5) + (-1, 3) = (2 + -1, -5 + 3) = (1, -2)$

2) Let $\vec{u} = (-4, 1)$ and $\vec{v} = (4, -1)$. Find $\vec{u} + \vec{v}$.

$\vec{u} + \vec{v} = (-4 + 4, 1 + -1) = (0, 0) = \vec{0}$

4. **Scalar Multiplication:** To multiply a vector $\vec{v}$ by a scalar c, multiply each of the components of $\vec{v}$ by c:

   $$c\vec{v} = c(v_1, v_2) = (cv_1, cv_2)$$

Ex:

3) Let $\vec{u} = (2, -1)$ and $\vec{v} = (4, 5)$, Find $\vec{w} = 3\vec{u} - \frac{1}{2}\vec{v}$. $= 3\vec{u} + \frac{-1}{2}\vec{v}$

$\vec{w} = 3(2, -1) + \frac{-1}{2}(4, 5) = (6, -3) + (-2, \frac{-5}{2})$

$\vec{w} = (6 - 2, -3 + \frac{-5}{2}) = (3, \frac{-11}{2})$

$-3 + \frac{-5}{2} = \frac{-6}{2} + \frac{-5}{2}$

$= \frac{-11}{2}$

The vector $-\vec{v} = (-1)\vec{v}$ is the *negative* of $\vec{v}$.

The difference of $\vec{u}$ and $\vec{v}$ is

$$\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$$ Additive Inverse

($\vec{v}$ is subtracted from $\vec{u}$ by adding its negative.)

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5. **Properties of Vector Addition and Scalar Multiplication:** Let $\vec{u}, \vec{v}$, and $\vec{w}$ be vectors in the plane and let c and d be scalars.

1) $\vec{u} + \vec{v}$ is a vector in the plane
2) $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
3) $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$
4) $\vec{u} + \vec{0} = \vec{u}$
5) $\vec{u} + (-\vec{u}) = \vec{0}$
6) $c\vec{u}$ is a vector in the plane
7) $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$
8) $(c + d)\vec{u} = c\vec{u} + d\vec{u}$
9) $c(d\vec{u}) = (cd)\vec{u}$
10) $1\vec{u} = \vec{u}$

6. **Vectors in $\mathbb{R}^n$:** Expanding from vectors in the plane to vectors in n-space, a vector is represented by an ordered n-tuple, which can be viewed as a point in $\mathbb{R}^n$ with the x_i's as coordinates, or as a vector with x_i's as its components. If $\vec{u}$ and $\vec{v}$ are vectors in $\mathbb{R}^n$ and c is a real number, then:

$$\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$
$$c\vec{u} = (cu_1, cu_2, \dots, cu_n)$$

(Properties are the same as with $\mathbb{R}^2$.)

$\mathbb{R}^n$ has the same properties as $\mathbb{R}^2$ (x, y)
$\mathbb{R}^3$ (x, y, z)
$(x_1, x_2, \dots, x_n)$

7. **Properties of Additive Identity and Additive Inverse:** Let $\vec{v}$ be a vector in $\mathbb{R}^n$ and c be a scalar. Then:

1) The additive identity (the vector $\vec{0}$) is unique, i.e if $\vec{v} + \vec{u} = \vec{v}$, then $\vec{u} = \vec{0}$.
2) $0\vec{v} = \vec{0}$
3) The additive inverse (the vector $-\vec{v}$) is unique; i.e. if $\vec{v} + \vec{u} = \vec{0}$, then $\vec{u} = -\vec{v}$.
4) $c\vec{0} = \vec{0}$
5) If $c\vec{v} = 0$, then $c = 0$ or $\vec{v} = \vec{0}$
6) $-(-\vec{v}) = \vec{v}$

Ex:

4) Let $\vec{u} = (0, -3, 1)$ and $\vec{v} = (-2, 1, 4)$. Find $\frac{1}{3}\vec{u} + 2\vec{v}$. $= \vec{\omega}$

$\vec{\omega} = \frac{1}{3}\vec{u} + 2\vec{v} = \frac{1}{3}(0, -3, 1) + 2(-2, 1, 4)$

$\vec{\omega} = (0, -1, \frac{1}{3}) + (-4, 2, 8)$

$\vec{\omega} = (-4, 1, \frac{25}{3})$

$\frac{1}{3} + 8 = \frac{1}{3} + \frac{24}{3}$
$= \frac{25}{3}$

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5) Write $\vec{x} = (-1, -2, -2)$ as a linear combination of $\vec{u} = (0,1,4)$, $\vec{v} = (-1,1,2)$, and $\vec{w} = (3, 1, 2)$.

What!?! - you need to find a, b, c so that

$$\vec{x} = a\vec{u} + b\vec{v} + c\vec{w}$$

$$(-1, -2, -2) = a(0,1,4) + b(-1,1,2) + c(3,1,2)$$

This means

$$-1 = a \cdot 0 + -b + 3c = -b + 3c$$
$$-2 = a + b + c$$
$$-2 = 4a + 2b + 2c$$

Solve using matrices

$$\begin{bmatrix} 0 & -1 & 3 & -1 \\ 1 & 1 & 1 & -2 \\ 4 & 2 & 2 & -2 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 1 & 1 & -2 \\ 0 & -1 & 3 & -1 \\ 4 & 2 & 2 & -2 \end{bmatrix} \xrightarrow{\substack{-R_2 \to R_2 \\ -4R_1 + R_3 \to R_3}}$$

$$\begin{bmatrix} 1 & 1 & 1 & -2 \\ 0 & 1 & -3 & 1 \\ 0 & -2 & -2 & 6 \end{bmatrix} \xrightarrow{\substack{-R_2 + R_1 \to R_1 \\ 2R_2 + R_3 \to R_3}} \begin{bmatrix} 1 & 0 & 4 & 3 \\ 0 & 1 & -3 & 1 \\ 0 & 0 & -8 & 8 \end{bmatrix} \xrightarrow{-\frac{1}{8}R_3 \to R_3}$$

$-4(1) + 2 = -2$ $\quad -4(1) + -2 = -6$ $\quad -4(-2) + -2 = 6$

$-(-3) + 1 = 4$ $\quad -(-2) + 1 = 3$

$2(1) + -2 = 0$ $\quad 2(-3) + -2 = -8$ $\quad 2(1) + 6 = 8$

$$\begin{bmatrix} 1 & 0 & 4 & 3 \\ 0 & 1 & -3 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \xrightarrow{\substack{-4R_3 + R_1 \to R_1 \\ 3R_3 + R_2 \to R_2}} \begin{bmatrix} a & b & c & \\ 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$a = 7$
$b = -2$
$c = -1$

$-4(-1) + 3 = 7$ $\quad 3(-1) + 1 = -2$

$$\vec{x} = 7\vec{u} - 2\vec{v} - \vec{w}$$

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4.2: Vector Spaces

MATH 2360

Definitions

1. **Vector Space:** Let $\mathcal{V}$ be a set on which vector addition and scalar multiplication are defined. If the following axioms are satisfied for every $\vec{u}$, $\vec{v}$, and $\vec{w}$ in $\mathcal{V}$ and every scalar c and d, then $\mathcal{V}$ is a vector space.

   1) $\vec{u}+\vec{v}$ is in $\mathcal{V}$ – Addition Rule
   2) $\vec{u}+\vec{v} = \vec{v}+\vec{u}$
   3) $\vec{u}+(\vec{v}+\vec{w}) = (\vec{u}+\vec{v})+\vec{w}$
   4) $\mathcal{V}$ has a zero vector $\vec{0}$ such that $\vec{u}+\vec{0} = \vec{u}$ for every $\vec{u}$ in $\mathcal{V}$
   5) For every $\vec{u}$ in $\mathcal{V}$, there is a vector $-\vec{u}$ in V such that $\vec{u}+(-\vec{u}) = \vec{0}$
   6) $c\vec{u}$ is in $\mathcal{V}$ Scalar multiplication
   7) $c(\vec{u}+\vec{v}) = c\vec{u}+c\vec{v}$
   8) $(c+d)\vec{u} = c\vec{u}+d\vec{u}$
   9) $c(d\vec{u}) = (cd)\vec{u}$
   10) $1\vec{u} = \vec{u}$

What does "closed" mean? ① The vector space must be nonempty (must have at least 1 element)

② The addition of 2 vectors must be in the vector space

③ Scalar multiplication must be in the vector space.

Examples of Vector Spaces

$\in$ means contained in

1) The set of all ordered pairs of real numbers with the standard operations. Form: $\vec{v}=(v_1, v_2)$

It is nonempty b/c $(0,0) \in \mathcal{V}$

? Is $\vec{u}+\vec{v}$ in $\mathcal{V}$? – yes b/c when you add to real #, you get a real #

? Is $c\vec{u} \in \mathcal{V}$? – yes b/c the product of a constant & $\vec{u}$ is also a real # and in $\mathcal{V}$.

2) The set of all ordered n-tuples of real numbers with the standard operations.

Form: $\vec{v}=(v_1, v_2, \ldots, v_n)$ b/c all v are $\mathbb{R}$. We know it is closed under both addition and scalar mult. We also know it is nonempty.

3) The set of all 4×2 matrices with matrix addition and scalar multiplication.

Form: $\vec{a} = A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \\ a_{41} & a_{42} \end{bmatrix}$

- The VS is nonempty b/c $\vec{0}$ matrix is in VS
- This is also closed under both addition and scalar multiplication.

4) The set of all polynomials of degree 2 or less with addition and scalar multiplication.

Form: $p(x)=a_0+a_1x+a_2x^2$ - This is nonempty b/c $1+x+2x^2$ in space

- The "or less" make it closed under addition
- It is closed under scalar mult b/c we can mult by const

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Important Vector Spaces Know

$\mathbb{R}^n$	ordered pairs in $\mathbb{R}$
$C(-\infty, \infty)$	all functions with domain $(-\infty, \infty)$
$C[a, b]$	all functions on a bounded interval $[a, b]$
P_n	all polynomials of the same power
$M_{m\times n}$	all matrices of the same size

$\mathbb{Q}$ - Rational Numbers (fraction or repeating/terminating decimals)

$\mathbb{Z}$ - Integers

$\mathbb{C}$ - complex Numbers $a \pm bi$

What do we need to make a vector space?

① nonempty
② closed under addition
③ closed under scalar multiplication

2. **Theorem:** Let $\vec{v}$ be any element of a vector space $\mathcal{V}$ and c be a scalar. Then:
   1) $0\vec{v} = \vec{0}$
   2) $c\vec{0} = \vec{0}$
   3) If $c\vec{v} = 0$, then $c = 0$ or $\vec{v} = \vec{0}$
   4) $(-1)\vec{v} = -\vec{v}$

Examples of Non-Vector Spaces – Must show 1 ex where it fails

1) The set of all integers with standard operations.

$\frac{1}{2}$ is a scalar $\quad \frac{1}{2}(1) = \frac{1}{2} \notin \mathbb{Z}$

2) The set of all 2nd degree polynomials with standard operations. – Must be second degree

$P_2 =$ $\quad x^2 \in P_2 \quad (1+x-x^2) \in P_2$ but $x^2+1+x-x^2 \notin P_2$
$= 1+x$ no square

3) The set of all ordered pairs with addition and $c(x_1, x_2) = (cx_1, 0)$.

$(1,1) = (x_1, x_2)$ $\quad V = \{x_1, x_2 \in \mathbb{R} \mid c(x_1,x_2) = (cx_1, 0)\}$

$c = 1$

$c(x_1, x_2) = 1(1,1) = (1,1) \neq (1\cdot 1, 0) = (1,0)$

not closed under scalar multiplication

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4.3: Subspaces of Vector Spaces

MATH 2360

⇒ then
→ implies
↔ iff - if and only if

Definitions - symbol $\subseteq$

1. **Subspace:** A nonempty subset $\mathcal{W}$ of a vector space $\mathcal{V}$ is a subspace of $\mathcal{V}$ when $\mathcal{W}$ is a vector space under the operations of addition and scalar multiplication defined in $\mathcal{V}$.

2. **Test for a Subspace:** If $\mathcal{W}$ is a nonempty subset of a vector space $\mathcal{V}$, then $\mathcal{W}$ is a subspace of $\mathcal{V}$ iff the two closure conditions hold:

   (1) If $\vec{u}$, $\vec{v}$ are in $\mathcal{W}$, then $\vec{u} + \vec{v}$ is in $\mathcal{W}$.

   (2) If $\vec{u}$ is in $\mathcal{W}$ and c is a scalar, then $c\vec{u}$ is in $\mathcal{W}$.

Show: ① Nonempty - there is at least 1 element
② Closed under addition - when 2 elem in VS added ⇒ ans in VS
③ Closed under scalar mult - when mult by any # ∈ ℝ ⇒ ans in VS

Give an example if not true

Example: Remember ℝ is closed & $\mathbb{R}^2$ is closed

1) Show that the set $\mathcal{W} = \{(x_1, 0, x_2): x_1, x_2 \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$ with the standard operations.

① $(0,0,0) \in \mathcal{W}$ b/c $0 \in \mathbb{R}$ ⇒ Nonempty

② $(x_1,0,x_2) + (y_1,0,y_2) = (x_1+y_1, 0, x_2+y_2)$, given $x_1, x_2, y_1, y_2 \in \mathbb{R}$ then we know $x_1+y_1, x_2+y_2 \in \mathbb{R}$ ⇒ closed under addition

③ $c(x_1,0,x_2) = (cx_1, 0, cx_2)$, given $c, x_1, x_2 \in \mathbb{R}$ then we know $cx_1, cx_2 \in \mathbb{R}$ ⇒ closed under scalar mult.

3. Every vector space has two trivial subspaces, $\mathcal{W} = \{\vec{0}\}$ (the **zero subspace**) and $\mathcal{W} = \mathcal{V}$.

4. Other subspaces are known as **proper** (nontrivial) subspaces – means all the elements are in the vector space, but not it is not the entire vector space.

Symbol $\subset$ b/c not = to entire space

Ex: → $A^T = A$

2) Show that the set of all 2×2 symmetric matrices is a subspace of $M_{2,2}$ with standard operations.

① $A, A^T \in M_{2,2}$ ⇒ nonempty → Given $cA_1, A_1, A_1^T \in M_{2,2}$ and symmetric
A_2, A_1^T

② $(A_1 + A_2)^T = A_1^T + A_2^T = A_1 + A_2$

⇒ Closed under addition

③ $c(A_1^T) = (cA_1)^T = cA_1$

⇒ Closed under scalar mult.

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→ do not have inverse $|A|=0$

3) Show that the set of all 2×2 singular matrices is not a subspace of $M_{2,2}$ with standard operations.

$A=\begin{bmatrix}0 & 2\\ 0 & 0\end{bmatrix}$, $|A|=0$, $B=\begin{bmatrix}0 & 0\\ -1 & 0\end{bmatrix}$ $|B|=0$ ⇒ $A, B \in$ subspace

$A+B=\begin{bmatrix}0 & 2\\ 0 & 0\end{bmatrix}+\begin{bmatrix}0 & 0\\ -1 & 0\end{bmatrix}=\begin{bmatrix}0 & 2\\ -1 & 0\end{bmatrix}$ $\quad |A+B|=\begin{vmatrix}0 & 2\\ -1 & 0\end{vmatrix}=0+2\neq 0$

⇒ $A+B$ is not in subsp

∴ singular matrices are not a subspace of $M_{2,2}$

4) Show that $\mathcal{W}=\{(x_1,x_2): x_1 \geq 0 \text{ and } x_2 \geq 0\}$ with standard operations is not a subspace of $\mathbb{R}^2$.

Consider the pt $(2,4)\in W$ and let $c=-2\in\mathbb{R}$

⇒ $-2(2,4)=(-4,-8)\notin W$

∴ W is not closed under scalar mult

5. **Intersection of Subspaces:** If $\mathcal{V}$ and $\mathcal{W}$ are both subspaces of vector space $\mathcal{U}$, then the intersection of $\mathcal{V}$ and $\mathcal{W}$ ($\mathcal{V} \cap \mathcal{W}$) is also a subspace of $\mathcal{U}$.

$V, W \subseteq U$ ⇒ $V \cap W \subseteq U$

Ex:

5) Determine whether each of the following is a subset of $\mathbb{R}^2$.

a) The set of points on the line $x + 2y = 0$. → $U=\{x+2y=0 : (-2x, x)\in\mathbb{R}^2\}$

① $(0,0)\in\mathbb{R}^2$ and $0+2(0)=0$ ⇒ U is non empty

② Let $\vec{u}=(-2t_1,t_1)$ and $\vec{v}=(-2t_2,t_2)$. $\vec{u},\vec{v}\in U$

⇒ $\vec{u}+\vec{v}=(-2t_1,t_1)+(-2t_2,t_2)=(-2(t_1+t_2),t_1+t_2)\in U$

⇒ Closed under addition

③ Let $\vec{u}=(-2t_1,t_1)\in U$ and $c\in\mathbb{R}$ ⇒ $c\vec{u}=c(-2t_1,t_1)$
$=(-2ct_1,ct_1)\in U$

⇒ Closed under scalar mult.

∴ $U\subseteq\mathbb{R}^2$

b) The set of points on $x^2 + y^2 = 1$. → $U = \{x^2+y^2=1 : (x,y) \in \mathbb{R}^2\}$

Consider $(1,0), (0,1) \in \mathbb{R}^2$

$(1,0)^2 + (0,1)^2 = (1,0) + (0,1) = (1,1) \neq 1$

not on $x^2+y^2=1$

$\therefore$ $x^2+y^2=1$ not a subspace

6) Determine whether each of the following is a subspace of $\mathbb{R}^3$.

a) $\mathcal{W} = \{(x_1, x_2, 1): x_1, x_2 \in \mathbb{R}\}$

$(0,0,0) \notin W$

$\Rightarrow W$ is not a subspace of $\mathbb{R}^3$

b) $\mathcal{W} = \{(x_1, x_1 + x_3, x_3); x_1, x_3 \in \mathbb{R}\}$

① $(0,0,0) \in W$ → let $x_1 = 0$, $x_3 = 0$ $\Rightarrow$ $x_1 + x_3 = 0$ $\Rightarrow$ Nonempty

② Let $\vec{u} = (u_1, u_1+u_3, u_3)$, $\vec{v} = (v_1, v_1+v_3, v_3) \in W$

Then

$\vec{u}+\vec{v} = (u_1, u_1+u_3, u_3) + (v_1, v_1+v_3, v_3) = (u_1+v_1, (u_1+v_1)+(u_3+v_3), u_3+v_3)$

$= (x_1, x_1+x_3, x_3) \in W$

$\Rightarrow$ Closed under addition

③ Let $c \in \mathbb{R}$, $\vec{u} = (u_1, u_1+u_3, u_3) \in W$. Then

$c\vec{u} = c(u_1, u_1+u_3, u_3) = (cu_1, c(u_1+u_3), cu_3)$

$= (cu_1, cu_1+cu_3, cu_3) = (x_1, x_1+x_3, x_3) \in W$

$\Rightarrow$ Closed under scalar mult

$\therefore$ $W \subseteq \mathbb{R}$

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4.4: Spanning Sets and Linear Independence

MATH 2360

Definitions

(lc)

1. **Linear Combination:** A vector $\vec{v}$ in a vector space $\mathcal{V}$ is a linear combination of vectors $\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_k$ in $\mathcal{V}$ when $\vec{v}$ can be written $\vec{v} = c_1\vec{u}_1 + c_2\vec{u}_2 + \cdots + c_n\vec{u}_n$, where $c_1, c_2, \ldots, c_n$ are scalars.

Examples:

1) Show that $\vec{v}_1 = (2,-1,1)$ is a linear combination of $\vec{v}_2 = (4,3,-1)$ and $\vec{v}_3 = (1,2,-1)$.

$\vec{v}_1 = a\vec{v}_2 + b\vec{v}_3 \rightarrow (2,-1,1) = a(4,3,-1) + b(1,2,-1) = (4a,3a,-a) + (b,2b,-b)$

$2 = 4a + b$
$-1 = 3a + 2b \Rightarrow \left[\begin{array}{cc|c} 4 & 1 & 2 \\ 3 & 2 & -1 \\ -1 & -1 & 1 \end{array}\right] -R_3 \rightarrow R_3 \left[\begin{array}{cc|c} 4 & 1 & 2 \\ 3 & 2 & -1 \\ 1 & 1 & -1 \end{array}\right] R_1 \Leftrightarrow R_2 \rightarrow \left[\begin{array}{cc|c} 1 & 1 & -1 \\ 3 & 2 & -1 \\ 4 & 1 & 2 \end{array}\right] \begin{array}{l} -3R_1+R_2 \rightarrow R_2 \\ -4R_1+R_3 \rightarrow R_3 \end{array} \left[\begin{array}{cc|c} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & -3 & 6 \end{array}\right]$
$1 = -a - b$

$R_2+R_1 \rightarrow R_1 \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{array}\right] -R_2 \rightarrow R_2 \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right] \Rightarrow (2,-1,1) = (4,3,-1) - 2(1,2,-1)$
$-3R_2+R_3 \rightarrow R_3$ $\quad a = 1$, $b = -2$

2) Write $\vec{u} = (1,1,1)$ and $\vec{v} = (1,-2,2)$ as a linear combination of the vectors in $\mathcal{S} = \{(1,2,3),(0,1,2),(-1,0,1)\}$.

a) $(1,1,1) = c_1(1,2,3) + c_2(0,1,2) + c_3(-1,0,1) \Rightarrow \begin{array}{l} c_1 - c_3 = 1 \\ 2c_1 + c_2 = 1 \\ 3c_1 + 2c_2 + c_3 = 1 \end{array} \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 1 \end{array}\right]$

$\begin{array}{l} -2R_1+R_2 \rightarrow R_2 \\ -3R_1+R_3 \rightarrow R_3 \end{array} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 2 & 4 & -2 \end{array}\right] -2R_2+R_3 \rightarrow R_3 \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$\Rightarrow c_1 - c_3 = 1$, $c_2 + 2c_3 = 1$ $\Rightarrow$ $c_3 = t$, $c_2 = 1-2t$, $c_1 = 1+t$ $\Rightarrow \vec{c} = \begin{bmatrix} 1+t \\ 1-2t \\ t \end{bmatrix}$ $\rightarrow$ Let $t=1$, $\vec{c} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ $\Rightarrow \vec{u} = 2(1,2,3) - (0,1,2) + (-1,0,1)$

b) $(1,-2,2) = c_1(1,2,3) + c_2(0,1,2) + c_3(-1,0,1)$

$\begin{array}{l} 1 = c_1 - c_3 \\ -2 = 2c_1 + c_2 \\ 2 = 3c_1 + 2c_2 + c_3 \end{array} \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & -2 \\ 3 & 2 & 1 & 2 \end{array}\right] \begin{array}{l} -2R_1+R_2 \rightarrow R_2 \\ -3R_1+R_3 \rightarrow R_3 \end{array} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 2 & 4 & 8 \end{array}\right] -2R_2+R_3 \rightarrow R_3 \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 0 & 0 & 16 \end{array}\right]$

No soln

Definitions

2. **Spanning Set:** Let $\mathcal{S} = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$ be a subset of a vector space $\mathcal{V}$. The set $\mathcal{S}$ is a **spanning** *set* of $\mathcal{V}$ when every vector in $\mathcal{V}$ can be written as a linear combination of vectors in $\mathcal{S}$.
   We say $\mathcal{S}$ spans $\mathcal{V}$.

Cr not need all

3b) $S = \{1, x, x^2\}$ spans P_2

Examples

3) Why does $\mathcal{S} = \{(1,0,0),(0,1,0),(0,0,1)\}$ spans $\mathbb{R}^3$? - Can S be written as lc for every vector in $\mathbb{R}^3$? Yes, let $c_1, c_2, c_3 \in \mathbb{R}$ and $(x,y,z) \in \mathbb{R}^3$

Then $c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1) = (x,y,z)$ is true for any value of c's.

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4) Why does $S = \{(1,2,3),(0,1,2),(-1,0,1)\}$ does not span $\mathbb{R}^3$? Let $\vec{u} = (u_1, u_2, u_3) \in \mathbb{R}^3$.

To span $\mathbb{R}^3$, $c_1(1,2,3) + c_2(0,1,2) + c_3(-2,0,1) = (u_1, u_2, u_3)$

$c_1 - 2c_3 = u_1$
$2c_1 + c_2 = u_2$
$3c_1 + 2c_2 + c_3 = u_3$

$$\Rightarrow \begin{vmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix} = 1\begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} - 0 + (-1)\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} = 1(1-0) - 1(4-3) = -1 \neq 0$$

$\Rightarrow$ unique soln $\therefore$ S spans $\mathbb{R}^3$

5) Show $S = \{(1,2,3),(0,1,2),(-1,0,1)\}$ spans $\mathbb{R}^3$.

does not

From earlier ex, we know $(1,-2,2)$ is not lc of vectors in S.

Therefore, S does not span $\mathbb{R}^3$

Definitions

3. **Span:** If $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is a set of vectors in a vector space $\mathcal{V}$, then the *span* of S is the set of all linear combinations of the vectors in S S, $\text{span}(S) = \text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\} = \{c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k\}$. When $\text{span}(S) = \mathcal{V}$, we say S spans $\mathcal{V}$, or $\mathcal{V}$ is spanned by S.

4. **Thm:** If $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is a set of vectors in a vector space $\mathcal{V}$, then $\text{span}(S)$ is a subspace of $\mathcal{V}$. Moreover, $\text{span}(S)$ is the smallest subspace of $\mathcal{V}$that contains S.

5. **Linear Independence:** A set of vectors $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ in a vector space $\mathcal{V}$ is **linearly independent** when $c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k = \vec{0}$ has only the trivial solution $c_1 = c_2 = \cdots = c_k = 0$. Otherwise, S is linearly dependent.

Test: linearly independ $\Rightarrow$ when $c\vec{v} = 0 \Rightarrow c = 0$

Notes: To test for linear independence, use matrices to determine if $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$ has more than the trivial solution.

Ex:

6) Determine whether the following sets of vectors are linearly independent.

a) $S = \{(1,2,3),(0,1,2),(-2,0,1)\}$

? $c_1(1,2,3) + c_2(0,1,2) + c_3(-2,0,1) = (0,0,0)$

$c_1 - 2c_3 = 0$
$2c_1 + c_2 = 0$
$3c_1 + 2c_2 + c_3 = 0$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{array}\right] \begin{matrix} \\ -2R_1 + R_2 \to R_2 \\ -3R_1 + R_3 \to R_3 \end{matrix} \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 2 & 7 & 0 \end{array}\right] \to -2R_2 + R_3 \to R_3$$

$-2(-2) + 0 = 4$
$-3(-2) + 1 = 7$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \begin{matrix} 2R_3 + R_1 \to R_1 \\ -4R_3 + R_2 \to R_2 \end{matrix} \to \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \Rightarrow \text{linearly independent}$$

b/c $c_1 = 0$ $c_2 = 0$ $c_3 = 0$

$-2(4) + 7 = 1$

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b) $S = \{1 + x - 2x^2,\ 2 + 5x - x^2,\ x + x^2\}$

? $c_1(1+x-2x^2) + c_2(2+5x-x^2) + c_3(x+x^2) = 0$

$c_1 + c_1x - 2c_1x^2 + 2c_2 + 5c_2x - c_2x^2 + c_3x + c_3x^2 = 0 = 0 + 0x + 0x^2$

$c_1 + 2c_2 + 0c_3 = 0$

$(c_1 + 5c_2 + 1c_3)x = 0x$

$(-2c_1 - c_2 + c_3)x^2 = 0x^2$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 1 & 5 & 1 & 0 \\ -2 & -1 & 1 & 0 \end{array}\right] \xrightarrow{\begin{array}{l} -R_1+R_2 \to R_2 \\ 2R_1+R_3 \to R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 3 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{-R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \text{linearly dependent}$$

b/c $c_1 + 2c_2 = 0$ & $c_2 + c_3 = 0$ } they depend on each other

c) $S = \left\{\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}\right\}$

? $c_1\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} + c_2\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} + c_3\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

To solve - use matrices → let each row be a position in the matrices of S and the columns be c_1, c_2, c_3 ⇒ Solve using row ops

$$\begin{array}{c} \\ a_{11} \\ a_{12} \\ a_{21} \\ a_{22} \end{array}\begin{array}{c} \begin{array}{cccc} c_1 & c_2 & c_3 & \text{soln} \end{array} \\ \left[\begin{array}{ccc|c} 2 & 3 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & 0 \end{array}\right] \end{array} \xrightarrow{\begin{array}{l} R_1 \leftrightarrow R_2 \\ R_3 \leftrightarrow R_4 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 2 & 3 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right] \xrightarrow{\begin{array}{l} -2R_1+R_2 \to R_2 \\ -R_1+R_3 \to R_3 \\ \frac{1}{2}R_3 \to R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{R_3 \leftrightarrow R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{-3R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{-R_3+R_4 \to R_4} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\Rightarrow c_1 = 0,\ c_2 = 0,\ c_3 = 0$

linearly independent b/c c's are zero

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Definitions

6. **Thm:** A set $S = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}, k \geq 2$ is linearly dependent if and only if at least one of the vectors $\vec{v}_i$ can be written as a linear combination of the other vectors in S. If one vector is a lc ⇒ linear dependent

7. **Corollary:** Two vectors in a vector space V are linearly dependent if and only if one is a multiple of the other.

Examples

7) Show that one of the vectors in $S = \{1 + x - 2x^2, 2 + 5x - x^2, x + x^2\}$ can be written as a linear combination of the others. Note: S is same as Example 6 part b.

$\vec{v}_1$ $\vec{v}_2$ $\vec{v}_3$

$$\begin{array}{c} x^2 \quad x \quad a \\ \left[\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ -1 & 5 & 2 & 0 \\ 1 & 1 & 0 & 0 \end{array}\right] \end{array} \xrightarrow{\text{row ops}} \begin{array}{c} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \\ c_1 \quad c_2 \quad c_3 \end{array}$$

$\Rightarrow c_1 + 2c_2 = 0 \Rightarrow c_1 = -2c_2 \Rightarrow c_1 = \frac{2}{3}c_3$

$3c_2 + c_3 = 0 \Rightarrow c_2 = -\frac{1}{3}c_3 \Rightarrow c_2 = -\frac{1}{3}c_3$

$c_3 = t \qquad c_3 = t$

$$\Rightarrow \vec{c} = \begin{bmatrix} 2/3 \\ -1/3 \\ 1 \end{bmatrix} t$$

⇒ Let $t = 3$ (any value - use 3 to get rid of frac.

$$\vec{c} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

$$\Rightarrow 2\vec{v}_1 - \vec{v}_2 + 3\vec{v}_3 = 0$$

$$2\vec{v}_1 + 3\vec{v}_3 = \vec{v}_2$$

$$\therefore 2 + 5x - x^2 = 2(1 + x - 2x^2) + 3(x + x^2)$$

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4.5: Basis and Dimension

MATH 2360

Definitions

1. **Basis:** A set of vectors $\mathcal{S} = \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_n}\}$ in a vector space $\mathcal{V}$ is a basis for $\mathcal{V}$ when
   1) $\mathcal{S}$ spans $\mathcal{V}$
   2) $\mathcal{S}$ is linearly independent.

   } Must show ① unique soln $|A| \neq 0$ ② only trivial soln for $C\vec{x} = \vec{0}$

Note (standard basis of R^n): $\vec{e}_1 = (1, 0, \ldots, 0), \vec{e_2} = (0,1,0, \ldots, 0), \ldots, \vec{e_n} = (0,0, \ldots, , 1)$

Examples

1) Show $\mathcal{S} = \{(1,0,0), (0,1,0), (0,0,1)\}$ is a basis for $\mathbb{R}^3$. → show lin ind.

① $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \neq 0 \Rightarrow$ unique soln

② $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \Rightarrow \begin{matrix} C_1 = 0 \\ C_2 = 0 \\ C_3 = 0 \end{matrix} \Rightarrow$ Only trivial soln

$\therefore S$ spans $\mathbb{R}^3$

$\Rightarrow S$ is basis for $\mathbb{R}^3$, actually S is standard basis of $\mathbb{R}^3$

2) Show $\mathcal{S} = \{(1,1), (1,-1)\}$ is a basis for $\mathbb{R}^2$.

① $\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -1 - 1 = -2 \neq 0 \Rightarrow S$ spans $\mathbb{R}^2$

② $\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & -1 & 0 \end{array}\right] -R_1 + R_2 \to R_2 \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & -2 & 0 \end{array}\right] -\frac{1}{2}R_2 \to R_2 \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 0 \end{array}\right] -R_2 + R_1 \to R_1$

$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right] \Rightarrow C_1 = C_2 = 0$ is only trivial soln $\Rightarrow$ lin ind

$\therefore S = \{(1,1), (1,-1)\}$ is basis for $\mathbb{R}^2$

3) Show $\mathcal{S} = \{1, x, x^2, x^3\}$ is a basis for P_3.

① $\begin{vmatrix} 1 & x & x^2 & x^3 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 1 - 0 = 1 \neq 0 \Rightarrow S$ spans P_3

$\therefore S = \{1, x, x^2, x^3\}$ is basis for P_3

② $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \Rightarrow C_1 = C_2 = C_3 = C_4 = 0 \Rightarrow$ lin ind only trivial soln

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$A^{-1} = \begin{bmatrix} -1 & 2 & -1 \\ 4 & -7 & 2 \\ -2 & 4 & -1 \end{bmatrix}$

Theorems

2. **Thm. Uniqueness of a Basis Representation:** If $S = \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_n}\}$ is a basis for vector space $\mathcal{V}$, then every vector in $\mathcal{V}$ can be written in one and only one way as a linear combination of vectors in S.

   Note: $S = \{(1,2,3),(0,1,2),(-2,0,1)\}$ from previous section

$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ -2 & 0 & 1 \end{bmatrix}$ ↑ work on extra sheet

Ex:

4) If $S = \{(1,2,3),(0,1,2),(-2,0,1)\}$, find a unique representation of $\vec{u}_1 = (0,1,1)$ and $\vec{u}_2 = (2,-2,-1)$ using the vectors in S. $\Rightarrow A\vec{c} = \vec{u} \Rightarrow \vec{c} = A^{-1}\vec{u}$ & unique

$A^{-1}\vec{u}_1 = \begin{bmatrix} -1 & 2 & -1 \\ 4 & -7 & 2 \\ -2 & 4 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0+2+-1 \\ 0+-7+2 \\ 0+4+-1 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \\ 3 \end{bmatrix} \Rightarrow \begin{matrix} c_1 = 1 \\ c_2 = -5 \\ c_3 = 3 \end{matrix}$ → lin indep

$\Rightarrow u_1 = 1(1,2,3) - 5(0,1,2) + 3(-2,0,1)$

$A^{-1}\vec{u}_2 = \begin{bmatrix} -1 & 2 & -1 \\ 4 & -7 & 2 \\ -2 & 4 & -1 \end{bmatrix}\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2+-4+-1 \\ 8++14+2 \\ -4+-8+-1 \end{bmatrix} = \begin{bmatrix} -7 \\ 24 \\ -13 \end{bmatrix} \Rightarrow \begin{matrix} c_1 = -7 \\ c_2 = 24 \\ c_3 = -13 \end{matrix}$ → lin indep

$\vec{u}_2 = -7(1,2,3) + 24(0,1,2) + -13(-2,0,1)$

Skip

3. **Bases and Linear Dependence Thm:** If $S = \{\vec{v_1}, \vec{v_2}, \ldots \vec{v_n}\}$ is a basis for a vector space $\mathcal{V}$, then every set containing more than n vectors in $\mathcal{V}$ is linearly dependent.

4. **Number of Vectors in a Basis Thm:** If a vector space $\mathcal{V}$ has one basis with n vectors, then every basis for $\mathcal{V}$ has n vectors.

① span $\Rightarrow A\vec{x} = \vec{u} \Rightarrow |A| \neq 0$

② lin indep $\Leftrightarrow A\vec{x} = \vec{0}$ w/ only trivial soln

Ex:

5) Determine whether S is a basis:

a) $S = \{(1,-1,1),(3,5,-2)\}$ for $\mathbb{R}^2$. 3 → Must have one more vector to span

w/ 3 elements, we need 3 vectors

Too few vectors to span.

b) $S = \{1 + x^2, 3 - x, 4 - 7x. 2x^3\}$ for P_3.

1 2 3 4 → only need 3 vectors

→ Too many vectors

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Notes

5. **Dimension:** If a vector space $\mathcal{V}$ has a basis consisting of n vectors, then n is the dimension of $\mathcal{V}$, denoted $\dim(\mathcal{V}) = n$. When $\mathcal{V}$ consists solely of the zero vector, the dimension is defined to be 0.

$\dim(\mathbb{R}^n) = n$ $\quad$ $\dim(P_n) = n+1$ $\quad$ $\dim(M_{m,n}) = mn$ $\quad$ $\dim(\textit{subspace of } \mathcal{V}) \leq n$

Examples

6) Find the dimension of the following subspaces of $\mathbb{R}^3$:

a) $\mathcal{W} = \{(d, c-d, c) : c, d \in \mathbb{R}\}$

use coefficients of c & d

$c(0,1,1) + d(1,-1,0) \Rightarrow S = \{(0,1,1),(1,-1,0)\}$ is span for $\mathcal{W}$

$\Rightarrow \dim(\mathcal{W}) = 2$ b/c 2 vectors

b) $\mathcal{W} = \{(2a, a, 0) : a \in \mathbb{R}\}$

$(2a, a, 0) = a(2,1,0) \Rightarrow \mathcal{W}$ is spanned by $S = \{(2,1,0)\}$

$\Rightarrow \dim(\mathcal{W}) = 1$

c) $\mathcal{W} = \textit{subspace spanned by } \{(-1,2,5),(2,0,3),(-5,2,-1)\} = S$ $\quad$ $\vec{v}_1$ $\vec{v}_2$ $\vec{v}_3$

$A\vec{v} = 0 \Rightarrow c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3 = \vec{0}$

$c_1(-1,2,5) + c_2(2,0,3) + c_3(-5,2,-1) = \vec{0}$

$$\begin{bmatrix} -1 & 2 & -5 & 0 \\ 2 & 0 & 2 & 0 \\ 5 & 3 & -1 & 0 \end{bmatrix} \begin{matrix} -R_1 \to R_1 \text{ new} \\ 2R_1 + R_2 \to R_2 \\ 5R_1 + R_3 \to R_3 \end{matrix} \begin{bmatrix} 1 & -2 & 5 & 0 \\ 0 & 4 & -8 & 0 \\ 0 & 13 & -26 & 0 \end{bmatrix} \begin{matrix} \frac{1}{4}R_2 \to R_2 \\ \frac{1}{13}R_3 \to R_3 \end{matrix} \begin{bmatrix} 1 & -2 & 5 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 1 & -2 & 0 \end{bmatrix} \begin{matrix} 2R_2 + R_1 \to R_1 \\ -R_2 + R_3 \to R_3 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{only many soln} \Rightarrow \text{Not indep} \Rightarrow S \text{ spans } \mathcal{W} \text{ but not basis}$$

7) Find the dimension of the subspace $\mathcal{W}$ of all symmetric matrices in $M_{2,2}$.

$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \Rightarrow a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

coefficient in matrix

$\Rightarrow S = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\}$ spans $\mathcal{W}$ and is lin ind

$\therefore S$ is basis of $\mathcal{W}$ = All sym 2x2 matrices

$\mathcal{W}$ / $\dim(S) = 3$

↳ b/c 3 ele in basis

6. **Basis via Independence or Span Thm:** Let $\mathcal{V}$ be a vector space of dimension n. Then $\mathcal{S} = \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_n}\} = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is a basis for $\mathcal{V}$ if:
   1) $\mathcal{S}$ is linearly independent, **OR**
   2) $\mathcal{S}$ spans $\mathcal{V}$.

$|A| \neq 0$ $\quad$ $C\vec{x} = \vec{0} \Rightarrow C = \vec{0}$ $\quad$ } Not both - one or other

Examples

8) Show $\mathcal{S} = \left\{\begin{bmatrix}2\\-3\\1\\4\end{bmatrix}, \begin{bmatrix}0\\1\\-2\\-2\end{bmatrix}, \begin{bmatrix}0\\0\\-5\\1\end{bmatrix}, \begin{bmatrix}0\\0\\0\\-1\end{bmatrix}\right\}$ is a basis for $M_{4,1}$. $\Rightarrow$ 1) Only trivial soln: $C\vec{x} = \vec{0}$

$$\begin{bmatrix}2&0&0&0&0\\-3&1&0&0&0\\1&-2&-5&0&0\\4&-2&1&-1&0\end{bmatrix} \xrightarrow{\frac{1}{2}R_1 \to R_1} \begin{bmatrix}1&0&0&0&0\\-3&1&0&0&0\\1&-2&-5&0&0\\4&-2&1&-1&0\end{bmatrix} \xrightarrow[-R_1+R_3 \to R_3]{3R_1+R_2 \to R_2} \begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-2&-5&0&0\\0&-2&1&-1&0\end{bmatrix}$$

Realize b/c Lower triangular: none of elements change but colm you are working with

$\therefore$ Only get trivial soln

$$\xrightarrow[2R_2+R_4 \to R_4]{2R_2+R_3 \to R_3} \begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&-5&0&0\\0&0&1&-1&0\end{bmatrix} \xrightarrow{-\frac{1}{5}R_3 \to R_3} \begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&1&-1&0\end{bmatrix} \xrightarrow{-R_3+R_4 \to R_4} \begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&-1&0\end{bmatrix} \xrightarrow{-R_4 \to R_4} \begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\end{bmatrix}$$

$\Rightarrow$ S spans $M_{4,1}$ & S lin ind $\quad$ $\therefore$ S is basis for $M_{4,1}$

OR

2) $|A| \neq 0$

$$\begin{vmatrix}2&0&0&0\\-3&1&0&0\\1&-2&-5&0\\4&-2&1&1\end{vmatrix} = 2(1)(-5)(1) = -10 \neq 0 \Rightarrow \text{S spans } M_{4,1}$$

b/c Lower triangular

Example 4 work

$$A=\begin{bmatrix}1&2&3\\0&1&2\\-2&0&1\end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc}1&2&3&1&0&0\\0&1&2&0&1&0\\-2&0&1&0&0&1\end{array}\right] \xrightarrow{2R_1+R_3\to R_3} \left[\begin{array}{ccc|ccc}1&2&3&1&0&0\\0&1&2&0&1&0\\0&4&7&2&0&1\end{array}\right] \xrightarrow{\substack{-2R_2+R_1\to R_1\\ -4R_2+R_3\to R_3}}$$

$2(2)+0=4 \qquad 2(3)+1=7$

$$\left[\begin{array}{ccc|ccc}1&0&-1&1&-2&0\\0&1&2&0&1&0\\0&0&-1&2&-4&1\end{array}\right] \xrightarrow{-R_3\to R_3} \left[\begin{array}{ccc|ccc}1&0&-1&1&-2&0\\0&1&2&0&1&0\\0&0&1&-2&4&-1\end{array}\right] \xrightarrow{\substack{R_3+R_1\to R_1\\ -2R_3+R_2\to R_2}}$$

$-2(2)+3=-1 \qquad -4(2)+7=-1$

$$\left[\begin{array}{ccc|ccc}1&0&0&-1&2&-1\\0&1&0&4&-7&2\\0&0&1&-2&4&-1\end{array}\right] \Rightarrow A^{-1}=\begin{bmatrix}-1&2&-1\\4&-7&2\\-2&4&-1\end{bmatrix}.$$

$-2(-2)+0=4 \qquad -2(4)+1=-7 \qquad -2(-1)+0=2$

$$\left[\begin{array}{ccc|ccc}1&2&3&1&0&0\\0&1&2&0&1&0\\-2&0&1&0&0&1\end{array}\right] \xrightarrow{2R_1+R_3\to R_3} \left[\begin{array}{ccc|ccc}1&2&3&1&0&0\\0&1&2&0&1&0\\0&4&7&2&0&1\end{array}\right] \xrightarrow{\substack{-2R_2+R_1\to R_1\\ -4R_2+R_3\to R_3}}$$

$2(2)+0=4 \qquad 2(3)+1=7$

$$\left[\begin{array}{ccc|ccc}1&0&-1&1&-2&0\\0&1&2&0&1&0\\0&0&-1&2&-4&1\end{array}\right] \xrightarrow{-R_3\to R_3} \left[\begin{array}{ccc|ccc}1&0&-1&1&-2&0\\0&1&2&0&1&0\\0&0&1&-2&4&-1\end{array}\right] \xrightarrow{\substack{R_3+R_1\to R_1\\ -2R_3+R_2\to R_2}}$$

$-2(2)+3=-1 \qquad -4(2)+7=-1$

$$\left[\begin{array}{ccc|ccc}1&0&0&-1&2&-1\\0&1&0&4&-7&2\\0&0&1&-2&4&-1\end{array}\right]$$

$-2+1=-1 \qquad 4+-2=2 \qquad -2(-2)+0=4 \qquad -2(4)+1=-7 \qquad -2(-1)+0=2$

$A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ set up A as columns not rows

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 & 0 & 1 \end{array}\right] \begin{matrix} \\ -2R_1+R_2 \to R_2 \\ -3R_1+R_3 \to R_3 \end{matrix} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 4 & -2 & 1 & 0 \\ 0 & 2 & 7 & -3 & 0 & 1 \end{array}\right] -2R_2+R_3 \to R_3 \rightarrow$$

$-2(-2)+0 = 4$

$-3(-2)+1 = 7$

$-2(4)+7 = -1$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 4 & -2 & 1 & 0 \\ 0 & 0 & -1 & 1 & -2 & 1 \end{array}\right] \begin{matrix} -2R_3+R_1 \to R_1 \\ 4R_3+R_2 \to R_2 \\ -R_3 \to R_3 \end{matrix} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 4 & -2 \\ 0 & 1 & 0 & 2 & -7 & 4 \\ 0 & 0 & 1 & -1 & 2 & -1 \end{array}\right]$$

$-2(-2)+-3 = 1$ $\quad -2(1)+0 = -2$

$4(1)+-2 = 2$

$4(-2)+1 = -7$

$4(1)+0 = 4$

$-2(1)+1 = -1$

$-2(-2)+0 = 4$

$-2(1)+0 = -2$

$$\Rightarrow A^{-1} = \begin{bmatrix} -1 & 4 & -2 \\ 2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix}$$

$$\Rightarrow A^{-1}\vec{u}_1 = \begin{bmatrix} -1 & 4 & -2 \\ 2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0+4+-2 \\ 0+-7+4 \\ 0+2+-1 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$$

$$A^{-1}\vec{u}_2 = \begin{bmatrix} -1 & 4 & -2 \\ 2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} = \begin{bmatrix} -2+-8+2 \\ 4+14+-4 \\ -2+-4+1 \end{bmatrix} = \begin{bmatrix} -8 \\ 14 \\ -5 \end{bmatrix}$$

These are lin ind to other ans.

Consider $\mathbb{R}^3 \Rightarrow (i, j, k) \in \mathbb{R}^3$

You know how to span $\mathbb{R}^3$ .. → Any pt in $\mathbb{R}^3$

Ex: span of $\mathbb{R}^3$

- $(2, 4, 5) = 2\vec{i} + 4\vec{j} + 5\vec{k}$ linear combo
- $\left(\frac{1}{2}, \frac{2}{3}, \frac{1}{5}\right) = \frac{1}{2}\vec{i} + \frac{2}{3}\vec{j} + \frac{1}{5}\vec{k}$ linear combo
- $(.4, .\bar{3}, .75) = .4\vec{i} + .\bar{3}\vec{j} + .75\vec{k}$ linear combo

· some not all

Basis of $\mathbb{R}^3$ → $\vec{i} = (1,0,0)$ $\vec{j} = (0,1,0)$ $\vec{k} = (0,0,1)$

↳ smallest (in size - # values) that spans $\mathbb{R}^3$

Ex: Taking photo of homework with your phone.

- Holding phone perfectly ⊥ to sheet is basis
- Holding phone at angle is span. ⇒ Phone using bottom corners of paper as basis and does row ops to get other 2 corners to make span.

4.6: Rank of a Matrix and Systems of Linear Equations

MATH 2360

Definitions:

1. **Row/Column Space:** If A is an $m \times n$ matrix,
   a) The **row space** of A is the subspace of $\mathbb{R}^n$ spanned by the row vectors of A,
   b) The **column space** of A is the subspace of $\mathbb{R}^m$ spanned by the column vectors of A.

↳ Matrix built by columns.

↳ Matrix built by rows. $A_{Row} = (A_{Column})^T$

2. **Thm:** If A and B are row-equivalent matrices, then their row spaces are equal.

3. **Thm (Basis for a Row Space):** If matrix A is row-equivalent to matrix B in row-echelon form, then the nonzero vectors of B form a basis for the row space of A.

↳ To find, use RREF. The basis is the rows that aren't zero.

Example:

1) Find a basis for the row space of A.

$$A = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & -2 \end{bmatrix} \begin{matrix} \\ \\ 3R_1+R_3 \to R_3 \\ -3R_1+R_4 \to R_4 \\ -2R_1+R_5 \to R_5 \end{matrix} \to \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 9 & 9 & 8 \\ 0 & -5 & -5 & -8 \\ 0 & -6 & -6 & -8 \end{bmatrix} \begin{matrix} \\ \\ -9R_2+R_3 \to R_3 \\ 5R_2+R_4 \to R_4 \\ 6R_2+R_5 \to R_5 \end{matrix} \to \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 8 \\ 0 & 0 & 0 & -8 \\ 0 & 0 & 0 & -8 \end{bmatrix} \begin{matrix} \\ \\ \frac{1}{8}R_3 \to R_3 \\ R_3+R_4 \to R_4 \\ R_3+R_5 \to R_5 \end{matrix}$$

$3(3)+0=9$, $-3(3)+4=-5$, $-2(3)+0=-6$

$3(1)+6=9$, $-3(1)+-2=-5$, $-2(1)+-4=-6$

$3(3)+-1=8$, $-3(3)+1=-8$, $-2(3)+-2=-8$

$$\begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow B = \{(1,3,1,3), (0,1,1,0), (0,0,0,1)\}$$

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2) Find a basis for the subspace of $\mathbb{R}^3$ spanned by S = {(−1, 2, 5), (3, 0, 3), (5, 1, 8)}.

$$\begin{bmatrix} -1 & 2 & 5 \\ 3 & 0 & 3 \\ 5 & 1 & 8 \end{bmatrix} \begin{matrix} -R_1 \to R_1 \\ 3R_1+R_2 \to R_2 \\ 5R_1+R_3 \to R_3 \end{matrix} \begin{bmatrix} 1 & -2 & -5 \\ 0 & 6 & 18 \\ 0 & 11 & 33 \end{bmatrix} \begin{matrix} \frac{1}{6}R_2 \to R_2 \\ \frac{1}{11}R_3 \to R_3 \end{matrix} \to \begin{bmatrix} 1 & -2 & -5 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{bmatrix} -R_2+R_3 \to R_3$$

$3(2)+0=6 \quad 3(5)+3=18$

$5(2)+0=11 \quad 5(5)+8=33$

$$\begin{bmatrix} 1 & -2 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow B = \{(1,-2,-5), (0,1,3)\}$$

$$A = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & 4 & -2 \end{bmatrix}$$

3) Find a basis for the column space of A in the first example. *(column space of A is the row space of A^T)*

$$A^T = \begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 3 & 1 & 0 & 4 & 0 \\ 1 & 1 & 6 & -2 & 4 \\ 3 & 0 & -1 & 1 & -2 \end{bmatrix} \Rightarrow B = \{(1,0,-3,3,2)^T, (0,1,9,-5,-6)^T, (0,0,1,-1,-1)^T\}$$

4. **Thm:** The row space and column space of an $m \times n$ matrix A have the same dimension. The dimension of the row (or column) space of a matrix A is the rank of A (denoted rank(A)).

Ex:

4) Find the rank of A.

$$A = \begin{bmatrix} 1 & -2 & 0 & 1 \\ 2 & 1 & 5 & -3 \\ 0 & 1 & 3 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 0 & 1 \\ 2 & 1 & 5 & -3 \\ 0 & 1 & 3 & 5 \end{bmatrix} -2R_1+R_2 \to R_2 \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 5 & 5 & -5 \\ 0 & 1 & 3 & 5 \end{bmatrix} \frac{1}{5}R_2 \to R_2 \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 3 & 5 \end{bmatrix}$$

$$-R_2+R_3 \to R_3 \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 2 & 6 \end{bmatrix} \frac{1}{2}R_3 \to R_3 \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$-2(-2)+1=5$

$-2(1)+-3=-5$

$-1+3=2$

$-(-1)+5=6$

$\Rightarrow B = \{(1,-2,0,1), (0,1,1,-1), (0,0,1,3)\}$

$\Rightarrow \text{rank}(A) = 3$

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5. **Thm (Solutions of a Homogeneous System):** If A is an $m \times n$ matrix, then the set of all solutions of the homogeneous system of linear equations $A\vec{x} = \vec{0}$ is a subspace of $\mathbb{R}^n$ called the nullspace of A (denoted $N(A)$). The dimension of the nullspace is the nullity of A.

$N(A) = \{\vec{x_h} \in \mathbb{R}^n : A\vec{x} = \vec{0}\}$ Soln space: $A = \{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0}\}$

$N(A)$ = space of vectors

Ex:

5) Find the null space of A. => Solve $A\vec{x} = \vec{0}$

$$A = \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 6 & -5 & 4 \\ 1 & 2 & 0 & 3 \end{bmatrix}$$

nullity$(A) = 2$ b/c 2 lin ind vectors in null space

$$\left[\begin{array}{cccc|c} 1 & 2 & -2 & 1 & 0 \\ 3 & 6 & -5 & 4 & 0 \\ 1 & 2 & 0 & 3 & 0 \end{array}\right] \xrightarrow{\substack{-3R_1+R_2 \to R_2 \\ -R_1+R_3 \to R_3}} \left[\begin{array}{cccc|c} 1 & 2 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 2 & 0 \end{array}\right] \xrightarrow{\substack{2R_2+R_1 \to R_1 \\ -2R_2+R_3 \to R_3}} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

x_1 x_2 x_3 x_4

$-3(2)+6=0$, $-2+2=0$; $-3(-2)+-5=1$, $-(-2)+0=2$; $-3(1)+4=1$, $-1+3=2$, $2(1)+1=3$

$x_1 + 2x_2 + 3x_4 = 0 \Rightarrow x_1 = -2x_2 - 3x_4$

$x_3 + x_4 = 0 \Rightarrow x_3 = -x_4$

Let $x_2 = \Delta$, $x_4 = t$ $\Rightarrow \vec{x} = \begin{bmatrix} -2\Delta - 3t \\ \Delta \\ -t \\ t \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}\Delta + \begin{bmatrix} -3 \\ 0 \\ -1 \\ 1 \end{bmatrix}t \Rightarrow N(A) = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}$ vectors — null space

6. **Thm:** If A is an $m \times n$ matrix of rank r, then the dimension of the null space is $n - r$, $(n = rank(A) + nullity(A))$

rank (A) = # vectors in basis

Ex: → Find basis → RREF (A)

6) Find the rank and nullity of A, and a subset of column vectors forming a basis for the column space of A.

$$A = \begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{bmatrix}$$

A is $3 \times 5 \Rightarrow n = 5$

rank$(A) = 3$

$\Rightarrow$ nullity$(A) = n - $ rank$(A) = 5 - 3 = 2$

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{bmatrix} \xrightarrow{\substack{-R_2 \to R_2 \\ 2R_1+R_3 \to R_3}} \begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 3 & -1 & -3 \\ 0 & -1 & -3 & 1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{bmatrix} \xrightarrow{\substack{R_2+R_3 \to R_3 \\ -3R_2+R_4 \to R_4}}$$

$-3(3)+9=0$, $-3(-1)=3$, $-3(-3)+-12=-3$

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 3 & -1 & -3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & -3 \end{bmatrix} \xrightarrow{\substack{R_3 \Leftrightarrow R_4 \\ \frac{1}{3}R_4 \to R_4}} \begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 3 & -1 & -3 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\substack{-R_3+R_1 \to R_1 \\ R_3+R_2 \to R_2}} \begin{bmatrix} 1 & 0 & -2 & 0 & 1 \\ 0 & 1 & 3 & 0 & -4 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

lin indep

tells us cols in A form Basis for A

$B = \{(1, 0, -2, 0)^T, (0, -1, -1, 3)^T, (1, 1, -1, 0)^T\}$

7. **Thm (Solutions of a Nonhomogeneous System):** If $\vec{x}_p$ is a particular solution of the nonhomogeneous system $A\vec{x} = \vec{b}$, then every solution of this system can be written $\vec{x} = \overrightarrow{x_p} + \overrightarrow{x_h}$, where $\overrightarrow{x_h}$ is a solution of the corresponding homogeneous equation $A\vec{x} = \vec{0}$.

Ex:

7) Find the set of all solution vectors of the system:

$$\left[\begin{array}{cccc|c} 1 & 0 & -2 & 1 & 5 \\ 3 & 1 & -5 & 0 & 8 \\ 1 & 2 & 0 & -5 & -9 \end{array}\right] \quad -3R_1+R_2\to R_2,\ -R_1+R_3\to R_3$$

$-3(-2)+-5=1$
$-3(0)+0=-3$
$-3(5)+8=-7$

$$\left[\begin{array}{cccc|c} 1 & 0 & -2 & 1 & 5 \\ 0 & 1 & 1 & -3 & -7 \\ 0 & 2 & 2 & -6 & -14 \end{array}\right] \quad -2R_2+R_3\to R_3 \qquad \left[\begin{array}{cccc|c} 1 & 0 & -2 & 1 & 5 \\ 0 & 1 & 1 & -3 & -7 \\ 0 & 0 & 0 & 0 & 0 \\ x & y & z & w & \text{soln} \end{array}\right]$$

$x+-2z+w=5$
$y+z+-3w=-7$
$\Rightarrow \quad x=2z-w+5$
$y=-z+3w-7$
$\Rightarrow$ Let $z=t$, $w=p$ $\Rightarrow$

$x=2t-p+5$
$y=-t+3p-7$
$z=t$
$w=p$

$$\Rightarrow \vec{x} = \begin{bmatrix} 2t-p+5 \\ -t+3p-7 \\ t \\ p \end{bmatrix} = \underbrace{\begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} t + \begin{bmatrix} -1 \\ 3 \\ 0 \\ 1 \end{bmatrix} p}_{\vec{x}_h \text{ - soln of homogeneous } (A\vec{x}=\vec{0})} + \underbrace{\begin{bmatrix} 5 \\ -7 \\ 0 \\ 0 \end{bmatrix}}_{\vec{x}_p \text{ - particular soln}}$$

8. **Thm (Solutions of a System of Linear Equations):** The system $A\vec{x} = \vec{b}$ is consistent iff $\vec{b}$ is in the column space of A.

↑ nonhomogeneous problem

9. **Equivalent Conditions for Square Matrices**: If A is an $n \times n$ matrix, the following are equivalent:
   1) A is invertible.
   2) $A\vec{x} = \vec{b}$ has a unique solution for any $n \times 1$ matrix $\vec{b}$.
   3) $A\vec{x} = \vec{0}$ has only the trivial solution.
   4) A is row-equivalent to I_n.
   5) $|A| \neq 0$.
   6) $Rank(A) = n$
   7) The n row vectors of A are linearly independent.
   8) The n column vectors of A are linearly independent.

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$$A^T = \begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 3 & 1 & 0 & 4 & 0 \\ 1 & 1 & 6 & -2 & -4 \\ 3 & 0 & -1 & 1 & -2 \end{bmatrix} \begin{matrix} \\ -3R_1+R_2 \to R_2 \\ -R_1+R_3 \to R_3 \\ -3R_1+R_4 \to R_4 \end{matrix} \to \begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 0 & 1 & 9 & -5 & -6 \\ 0 & 1 & 9 & -5 & -6 \\ 0 & 0 & 8 & -8 & -8 \end{bmatrix} \begin{matrix} \\ \\ -R_2+R_3 \to R_3 \\ \frac{1}{8}R_3 \to R_3 \end{matrix}$$

$-3(-3)+0=9$
$-(-3)+6=9$
$-3(-3)+-1=8$

$-3(3)+4=-5$
$-(3)+-2=-5$
$-3(3)+1=-8$

$-3(2)+0=-6$
$-(2)+4=2$
$-3(2)+-2=-8$

$-(-6)+2=8$

$$\begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 0 & 1 & 9 & -5 & -6 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} R_3 \leftrightarrow R_4 \begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 0 & 1 & 9 & -5 & -6 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow B = \{(1,0,-3,3,2)^T, (0,1,9,-5,-6)^T, (0,0,1,-1,-1)^T\}$

B/c columns matrix

4.7: Coordinates and Change of Basis

MATH 2360

Definitions:

1. **Coordinate Representation Relative to a Basis:** Let $B = \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_n}\}$ be an ordered basis for vector space V and let $\vec{x}$ be a vector in V such that $\vec{x} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n$. Then the scalars $c_1, \ldots, c_n$ are the coordinates of $\vec{x}$ relative to basis B, and the coordinate matrix (vector) of $\vec{x}$ relative to B is the column matrix in $\mathbb{R}^n$ whose components are the coordinates of $\vec{x}$.

$$[\vec{x}_B] = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \qquad [\vec{x}_S] = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$ } coordinate relative to standard basis

Examples:

1) Find the coordinate matrix of $\vec{x} = (4, -1, 2)$ in $\mathbb{R}^3$ relative to the standard basis.

standard basis of $\mathbb{R}^3 = \{(1,0,0), (0,1,0), (0,0,1)\}$

$\Rightarrow \vec{x} = 4(1,0,0) + -1(0,1,0) + 2(0,0,1)$

$\Rightarrow [\vec{x}_B] = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$

2) The coordinate matrix of $\vec{x}$ in $\mathbb{R}^2$ relative to basis B = $\{(1,0), (1,2)\}$ is $\vec{x_B} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. Find the coordinate matrix of $\vec{x}$ relative to the standard basis.

standard basis: $B = \{(1,0), (0,1)\}$

$\vec{v}_1$ $\vec{v}_2$

$\vec{x} = 3(1,0) + 2(1,2) = (3,0) + (2,4) = (5,4) = 5(1,0) + 4(0,1)$

$\Rightarrow [\vec{x}_S] = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$

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→ stand basis: $B_S = \{(1,0,0),(0,1,0),(0,0,1)\}$

3) Find the coordinate matrix of $\vec{x} = (1,2,-1)$ in $\mathbb{R}^3$ relative to $B = \{(1,0,1),(0,-1,2),(2,3,-5)\}$.

$\vec{v}_1$ $\vec{v}_2$ $\vec{v}_3$

$\Rightarrow \vec{x} = c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3$

$\Rightarrow (1,2,-1) = c_1(1,0,1) + c_2(0,-1,2) + c_3(2,3,-5)$

$1 = c_1 + 2c_3$
$2 = -c_2 + 3c_3$
$-1 = c_1 + 2c_2 - 5c_3$

$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -1 & 3 & 2 \\ 1 & 2 & -5 & -1 \end{array}\right] -R_1+R_3 \to R_3 \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -1 & 3 & 2 \\ 0 & 2 & -7 & -2 \end{array}\right] -R_2 \to R_2,\ 2R_2+R_3 \to R_3$

$2(2)+-2=2$ $\quad 2(3)+-7=-1$ $\quad 2(2)+1=5$ $\quad -3(2)+-2=-8$

$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & -1 & 2 \end{array}\right] \begin{array}{l} 2R_3+R_1 \to R_1 \\ -3R_3+R_2 \to R_2 \\ -R_3 \to R_3 \end{array} \rightarrow \left[\begin{array}{ccc|c} c_1 & c_2 & c_3 & \\ 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -8 \\ 0 & 0 & 1 & -2 \end{array}\right]$

$\Rightarrow \vec{x} = 5(1,0,1) + -8(0,-1,2) + -2(2,3,-5)$

$\Rightarrow [\vec{x}]_B = \begin{bmatrix} 5 \\ -8 \\ -2 \end{bmatrix}$

2. **Change of Basis:** The procedure of finding coordinates relative to a basis B', given the coordinates relative to some basis B ($B \to B'$), is known as a change of basis. If we write $P\vec{x}_{B'} = \vec{x}_B$, P is the transition matrix from B' to B.

3. **Thm:** If P is the transition matrix from basis B' to basis B in $\mathbb{R}^n$, then P is invertible, and the transition matrix from B to B' is P^{-1}.

4. **Lemma:** Let $B = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ and $B' = \{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_n\}$ be two bases for a vector space V. If

$$\begin{aligned} \vec{v}_1 &= c_{11}\vec{u}_1 + c_{21}\vec{u}_2 + \cdots + c_{n1}\vec{u}_n \\ \vec{v}_2 &= c_{12}\vec{u}_1 + c_{22}\vec{u}_2 + \cdots + c_{n2}\vec{u}_n \\ &\vdots \\ \vec{v}_n &= c_{1n}\vec{u}_1 + c_{2n}\vec{u}_2 + \cdots + c_{nn}\vec{u}_n \end{aligned}$$

then the transition matrix from B to B' is

$$\begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix} = P^{-1}.$$

5. **Thm:** Transition Matrix from B to B'. Let $B = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ and $B' = \{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_n\}$ be two bases for $\mathbb{R}^n$. Then the transition matrix P^{-1} from B to B' can be found using Gauss-Jordan elimination to change $[B' \mid B]$ to the form $[I_n \mid P^{-1}]$. (RREF)

$[\text{new basis} \mid \text{basis}] \xrightarrow{\text{row ops}} [I \mid P^{-1}]$

↑ original basis

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Examples

4) Find the transition matrix from B to B' for the bases $B = \{(1,0,0),(0,1,0),(0,0,1)\}$ and $B' = \{(1,0,1),(0,1,2),(2,3,-5)\}$ in $\mathbb{R}^3$.

cols

$[B' \mid B] \xrightarrow{\text{row ops}} [I \mid P^{-1}]$

cols

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 1 & 2 & -5 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} -R_2 \to R_2 \\ -R_1 + R_3 \to R_3 \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & -1 & 0 \\ 0 & 2 & -7 & -1 & 0 & 1 \end{array}\right] -2R_2 + R_3 \to R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & -1 & 0 \\ 0 & 0 & -1 & -1 & 2 & 1 \end{array}\right] \begin{array}{l} 2R_3 + R_1 \to R_1 \\ -3R_3 + R_2 \to R_2 \\ -R_3 \to R_3 \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 4 & 2 \\ 0 & 1 & 0 & 3 & -7 & -3 \\ 0 & 0 & 1 & 1 & -2 & -1 \end{array}\right]$$

$2(-1)+1=-1 \quad -3(-1)+0=3$

$2(2)+0=4 \quad -3(2)+-1=-7$

$2(1)+0=2 \quad -3(1)+0=-3$

$$P^{-1} = \begin{bmatrix} -1 & 4 & 2 \\ 3 & -7 & -3 \\ 1 & -2 & -1 \end{bmatrix}$$

Bc to find A^{-1}: $[A \mid I] \xrightarrow{\text{row ops}} [I \mid A^{-1}]$

(Note that if B' is the standard basis, $P^{-1} = B$ and if B is the standard basis, $P^{-1} = (B')^{-1}$.)

5) Find the transition matrix from B to B' for $B = \{(-3,2),(4,-2)\}$ and $B' = \{(-1,2),(2,-2)\}$ in $\mathbb{R}^2$.

$[B' \mid B] \xrightarrow{\text{row ops}} [I \mid P^{-1}]$

$$\left[\begin{array}{cc|cc} -1 & 2 & -3 & 4 \\ 2 & -2 & 2 & -2 \end{array}\right] \begin{array}{l} -R_1 \to R_1 \\ \frac{1}{2}R_2 \to R_2 \end{array} \left[\begin{array}{cc|cc} 1 & -2 & 3 & -4 \\ 1 & -1 & 1 & -1 \end{array}\right] -R_1 + R_2 \to R_2 \left[\begin{array}{cc|cc} 1 & -2 & 3 & -4 \\ 0 & 1 & -2 & 3 \end{array}\right] 2R_2 + R_1 \to R_1$$

$$\left[\begin{array}{cc|cc} 1 & 0 & -1 & 2 \\ 0 & 1 & -2 & 3 \end{array}\right] \Rightarrow P^{-1} = \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix}$$

$2(-2)+3=-1$

$2(3)+-4=2$

6. **Coordinate Representation in n-Dimensional Spaces:** Coordinate representation allows us to represent vectors in any n-dimensional space using the same notation used in $\mathbb{R}^n$.

Examples

6) Find the coordinate matrix of $p = 4 - 2x^2 + 3x^3$ relative to the standard basis $S = \{1, x, x^2, x^3\}$ (for P_3),

$p = 4 - 2x^2 + 3x^3 = 4 + 0x - 2x^2 + 3x^3$

$$[p]_S = \begin{bmatrix} 4 \\ 0 \\ -2 \\ 3 \end{bmatrix} \leftarrow \text{coefficients of } p$$

7) Find the coordinate matrix of $\vec{X} = \begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix}$ relative to the standard basis $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$, for $M_{3,1}$.

means

$$\vec{x} = \begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix} = (-1)\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 4\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow [\vec{x}]_S = \begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix}$$

↑ Coordinate vector relative to S

↑ basicly const mult by standard basis